

9708/31/o/n/21 qp

1 – A:

A - Pareto efficiency, also known as Pareto optimality, is a state where resources are allocated in such a way that it is impossible to make one person better off without making someone else worse off. **“Pareto efficiency” is achieved when we are operating on the production possibility curve.** This means that the allocation of resources is optimal, and any changes would lead to a loss for at least one person.

B) Producing greater output with the same inputs is related to achieving productive efficiency.

C) If there is spare capacity in production, this represents a wastage of resources, which indicates that efficiency is not being achieved.

D) When resources can still be re-allocated and total satisfaction can be increased, this represents that we are operating inside of the PPC, which indicates that we are “pareto inefficient”

2 – D:

A – Non-provision of public goods by the market as a result of the “free-rider” problem results in market failure, but is not an example of a negative externality.

B – Indirect tax will fail to internalize the negative externality when the per-unit indirect tax is not equal to the size of the external cost, and in that case, the policy will be ineffective. This represents an ineffective policy rather than the existence of external costs.

C – When a monopoly power results in a reduced output and a higher price, this represents “under-allocation” being done by a monopoly. This is incorrect because again this is not an example of a negative externality.

D – When the production of a good has a harmful effect on the third party, who is not directly involved in the transaction, this represents “external costs”, as a result of which, this becomes correct.

3 – B:

A) This change is likely to support the government's aim of increasing allocative and productive efficiency. With cheaper loans, new businesses can enter the market, increasing competition leading to better resource allocation and productivity gains.

B) This change makes it LESS LIKELY that the government will meet its aim. Mergers and takeovers can lead to market concentration, reducing competition and potentially creating monopolies. This can stifle innovation and efficient resource allocation, undermining the government's objectives.

C) This change supports the government's aim. With more price information available, consumers can make informed decisions, driving businesses to improve efficiency and reduce prices.

D) This change supports the government's aim. Lower fixed costs can encourage new entrants, increase competition, and drive efficiency improvements as businesses strive to minimize costs.

4 – A:

When a rational consumer maximizes satisfaction by buying three units of good X and four units of good Y, it indicates that the consumer has **equalized the marginal utility per dollar spent on both goods. Since this represents that the consumer equilibrium where $MUX/PX = MUY/PY$ has been established, option A becomes correct.**

B) The total utility of 3X is not necessarily equal to the total utility of 4Y. The consumer has maximized satisfaction by allocating their budget in a way that equalizes the marginal utility per dollar spent, **not the total utility.**

C) The consumer has already maximized utility by buying three units of good X and four units of good Y. **Operating at the consumer equilibrium represents that satisfaction cannot be further maximized.** Buying one more unit of good X would not increase overall utility, as it would disrupt the optimal allocation.

D) The fact that the consumer buys more units of good Y (four) than good X (three) does not necessarily mean that good Y has greater marginal utility than good X.

5-B:

A – The amount that a household has available to spend does not represent a budget line on the indifference curve.

B - A budget line shows all possible combinations of two goods available to consumers that can be purchased with their money income . The cost of each of the combinations is equal to the given money income of the consumer, since operating on the “budget line” means that all income is spent. This is why, option B becomes correct.

C – The difference between a government’s income and expenditure represents a “fiscal budget”

D – The relationship between the price of the good and the qty demanded represents “the law of demand”, which is illustrated through the “demand curve”

6-D:

A: Horizontal integration occurs when two firms operating in the same industry, and at the same stage of the production integrate.

B: A cartel is **a formal agreement between a group of producers of a good or service in an oligopoly market structure to control supply or to fix prices.**

C- Vertical integration backwards occurs when the firm in the secondary sector merges with the firm in the primary sector. This could have occurred when the car-producer would have merged with supplier of rubber, for example.

D – Vertical integration forward occurs when the firm in the secondary sector merges with the one in the tertiary sector, or in other words, a firm merges with another firm operating in the same industry, but at a stage of production ahead of it. In this case, a car manufacturer merging with a car- retailer is an example of vertically-forward integration.

7-B:

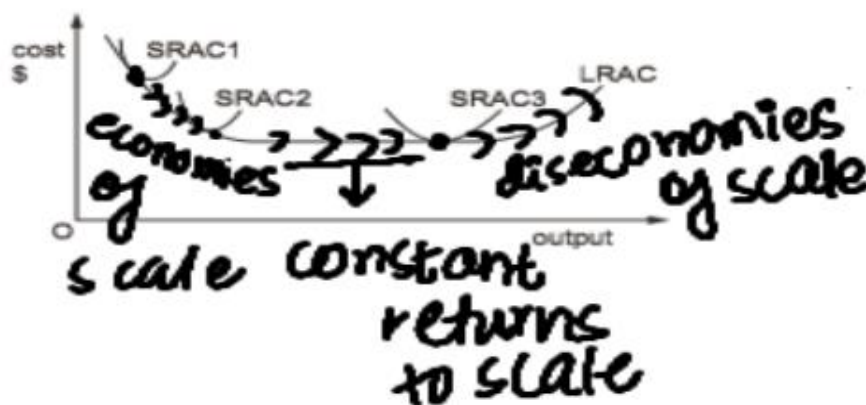
A) A monopoly is a market structure where a single seller dominates the market, supplying the entire market demand. However, in the scenario described, there are many sellers of hot foods competing for sales, which rules out the possibility of a monopoly.

B) Monopolistic competition is a market structure characterized by many firms producing differentiated products. The key feature is product differentiation, which allows firms to have some market power due to their unique offerings. In the hot food market, vendors sell similar but not identical products, **indicating product differentiation**. Each vendor may have a unique recipe, cooking style, or brand identity that distinguishes their product from others. This differentiation allows vendors to **engage in non-price competition, i.e.** compete on factors beyond price, such as taste, quality, and convenience. This makes option B correct.

C) An oligopoly is a market structure where a small number of firms compete with each other. In the hot food market, there are many vendors, not just a few, which rules out oligopoly. With many vendors, the market is more competitive.

D) Perfect competition is a market structure characterized by many firms producing a homogeneous product, with free entry and exit, and perfect information. In perfect competition, firms are price-takers, and there is no product differentiation. However, in the hot food market, vendors sell similar but **not identical products**, indicating some degree of **product differentiation**.

8 – C:



A – When the long-run average cost curve is horizontal, it represents “constant returns to scale” as can be seen above

B – When the long-run average cost curve is U-shaped, it represents both: economies of scale and dis-economies of scale

C – When the LRAC curve slopes downwards, it represents that there is a fall in Long run average costs, and in turn shows “economies of scale”. This can be seen above.

D - When the LRAC curve slopes upwards , it represents that there is a rise in Long run average costs, and in turn shows “dis-economies of scale”. This can be seen above.

9- C:

A – If a firm has a near monopoly status, and has control over prices, it can demonstrate producer sovereignty, and raise prices by a relatively higher percentage, which will contract qty demanded by a smaller percentage due to inelastic demand, resulting in a rise in revenues, and in turn profits.

B – If a firm locates in an area with a low level of corporation tax, it will face a fall in costs, and in turn a rise in profits.

C - Cartel occurs when two or more firms in an oligopoly openly merge. When there are production quotas, there is a restriction in increasing output, as a result of which, costs cannot be minimized. **Therefore, this would not help in achieving the objectives.**

D – If subsidiary firms are owned in different countries to achieve economies of scale, Average total costs will decline, and therefore, profits will rise.

10-B:

➔ Low price-elasticity of demand:

As a producer of luxury goods, a low Price Elasticity of Demand or **inelastic demand** is preferred. This means that demand is less sensitive to price changes, allowing the producer to increase prices without significantly reducing demand. As a result, the producer can maintain profit margins and revenue.

➔ High income-elasticity of demand:

As a producer of luxury goods, a high Income Elasticity of Demand is preferred. This means that as income increases, demand for the luxury good increases disproportionately, leading to a significant rise in sales and revenue. This is ideal for luxury goods, which are often purchased by high-income individuals who are willing to spend more on exclusive products.

11- B:

AR (Average Revenue) curve shifts right:

- With increased demand, the price or average revenue (AR) also increases.
- The AR curve shifts right, indicating that the firm's average revenue has increased.

MR (Marginal Revenue) curve shifts right:

Since the AR curve shifts towards the right, the MR curve shifts towards the right as well.

The MR curve shifts right, indicating that the firm can generate more revenue from each additional unit sold.

MC (Marginal Cost) curve remains unchanged:

- The advertising campaign does not affect the firm's marginal cost of production.
- The MC curve remains unchanged, as the cost of producing each additional unit remains the same.
This is because, advertising costs represent a fixed cost, which does not change when additional units are produced.

AC (Average Cost) curve shifts up:

- The \$100 million advertising campaign increases the firm's fixed costs.
- The average cost (AC) curve shifts up, indicating that the firm's average cost per unit has increased due to the additional fixed cost.

12 – D:

A – When a firm finances its new investment using retained profits, this represents that it will prefer to maximize profits so that it has sufficient retained profits for re-investment purposes.

B) Active shareholders may actually push for profit maximization since the dividend payouts of shareholders are linked to profits.

C) When investment banks make takeovers easy to implement, there is going to be an increase in the number of large firms due to an increase in takeover activity, which means that these large firms may actually aim for profit maximization.

D) This is known as the **principal-agent** (separation of ownership and control) problem. When a firm has many shareholders and is managed by employees (i.e., not the owners), the managers may prioritize their own interests over profit maximization. They may prefer sales revenue maximization as an objective that is achieved when $MR=0$ because their salaries and bonuses are linked to revenues.

13- C:

- **Increase Direct Tax:** Direct taxes, such as income taxes, are progressive, meaning higher-income individuals pay a larger share of their income in taxes. Increasing direct taxes reduces income inequality by redistributing wealth from the rich to the poor.
- **Decrease Indirect Tax:** Indirect taxes, such as sales taxes, are regressive, meaning lower-income individuals pay a larger share of their income in taxes. Decreasing indirect taxes reduces the burden on lower-income individuals and helps reduce income inequality.

- **Increase Availability of Benefits:** Increasing **means-tested** benefits, such as social security, unemployment benefits, and welfare programs, helps support lower-income individuals and reduces income inequality.

14-D:

A) This statement is incorrect. **Pollution permits** instead provide a continued incentive for pollution reduction.

B) In case of standards, the government sets specific limits or targets. In contrast, tradeable pollution permits create a market for pollution rights, allowing firms to buy and sell permits, which harnesses the market mechanism to reduce pollution.

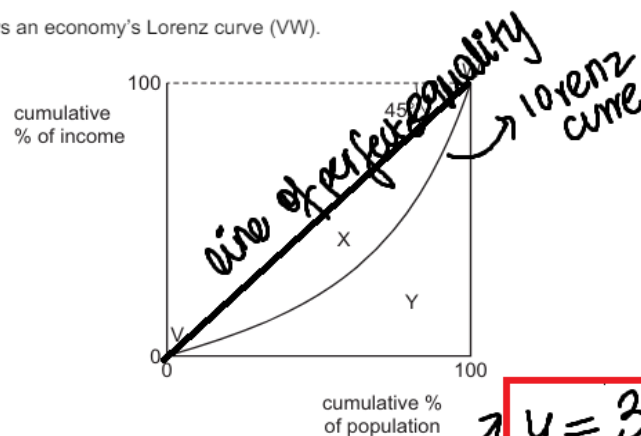
C) This statement is incorrect. Tradeable pollution permits require a complex system for tracking and trading permits, which can be difficult to establish and manage. In contrast, standards can be simpler to implement and enforce.

D) This statement is correct. Tradeable pollution permits treat all firms equitably, as they all have the same opportunity to buy or sell permits to meet their pollution targets. In contrast, standards can disproportionately affect certain firms, depending on their size, industry, or other factors. This explains why key D is correct.

15-B:

Gini-coefficient is measured by dividing the area between the line of perfect equality and the Lorenz curve denoted through X, with the total area under the line of perfect equality that is denoted through X+Y

The diagram shows an economy's Lorenz curve (VW).



The area of Y is three times bigger than the area of X.

What is the value of the economy's Gini coefficient?

- A $\frac{1}{3}$ **B $\frac{1}{4}$** C $\frac{3}{4}$ D $\frac{4}{3}$

$\Rightarrow$ gini coefficient =

$$\frac{X}{X+Y}$$

Substitute this

$$\frac{1X}{X+3X}$$

$$\frac{1X}{4X}$$

$$\Rightarrow \frac{1}{4} \rightarrow \text{B}$$

16-D:

A- An increase in marginal productivity of labor results in a rise in demand for labor, causing the demand for labor curve to shift rightwards.

B- An increase in the demand for product will result in a rise in demand for labor, which is a derived demand, causing the demand for labor curve to shift rightwards.

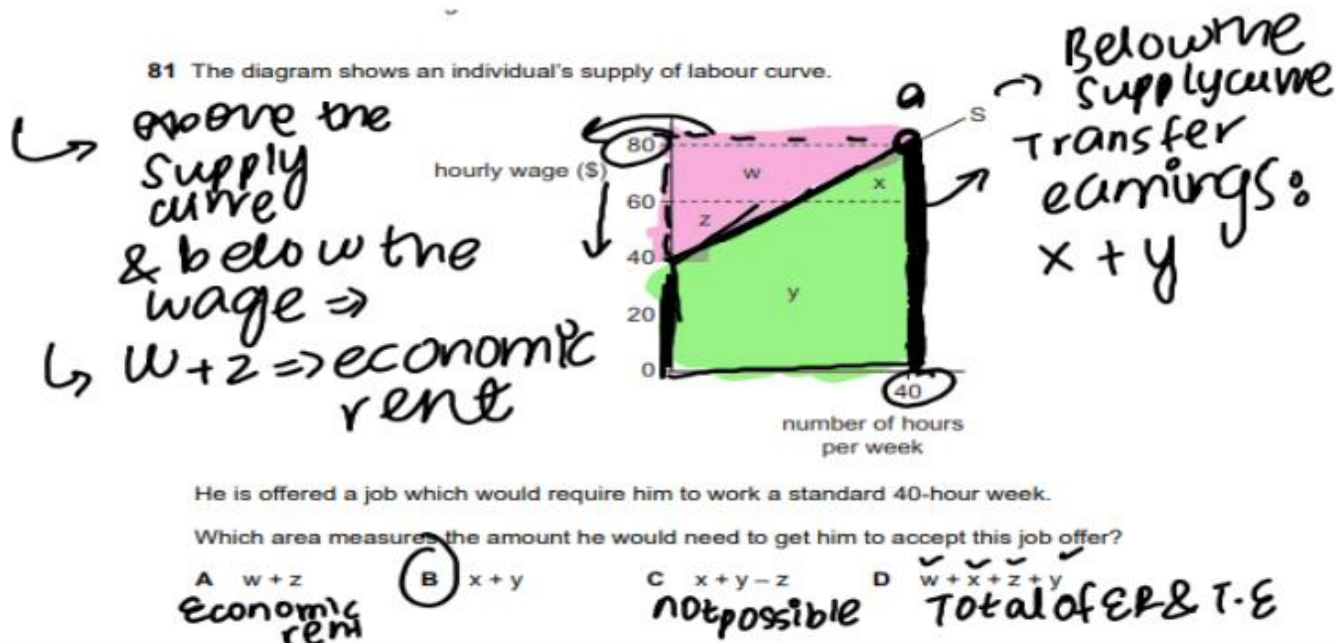
C- An increase in the price of the product as a result of a rise in the demand for the product will result in a rise in demand for labor, which is a derived demand, causing the demand for labor curve to shift rightwards.

D- This is correct because an increase in the wage rate causes movement along the demand curve instead of bringing a shift.

17-B:

Transfer earnings are the minimum income a worker needs in order to supply their labor.

The amount he would need to get him accept the job offer represents the "transfer earnings". Transfer earnings are denoted through the area below the supply curve.



This is why key B is correct.

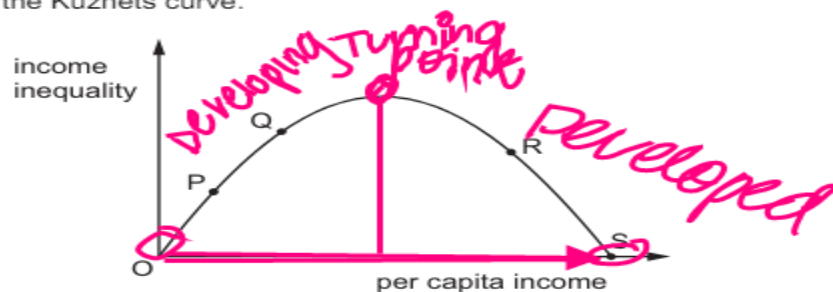
18 – D:

- 1) Country Y is “an open economy”. This is because, it is open to international trade, which can be shown through the fact that the equation of Aggregate demand of this country features “net exports” (X-M)
- 2) Moreover, the equation of AD does not include government spending (G), because of which, this means that country X is “without a government”

19-D:

7

The diagram shows the Kuznets curve.



- A- We cannot deduce if country P is a market economy through the **Kuznets curve**. This is because, the Kuznets curve shows the relationship between income inequality and per capita income overtime as the economy develops.
- B- According to the Kuznets curve, country Q is a developing economy. This can be seen in the graph above.
- C- According to the Kuznets curve, country R is a developed economy .This can be seen in the graph above.
- D- This is correct to say that country S has a gini coefficient of 0

20-A:

The correct answer is A) They want to avoid double counting.

Value added is the difference between the value of a firm's output and the value of the intermediate goods and services used to produce that output. By using value added, economists can avoid double counting, which occurs when the same output is counted multiple times in the production process.

For example, consider a car manufacturer that buys tires from a tire manufacturer. If the value of the tires is included in the value of the car, and the tire manufacturer's output is also included in the total output, then the value of the tires would be counted twice. By using value added, the value of the tires is only counted once, avoiding double counting.

Wheat Farmer's Contribution to GDP

Output: 100 kg of wheat worth \$100

Intermediate input: \$40 worth of seeds (already produced by the seed industry)

To calculate the farmer's contribution to GDP, we need to find the value added by the farmer to the crop.

- Value added = Output - Intermediate input
= \$100 - \$40
= \$60

This means the farmer has added \$60 of value to the crop through their labor, land, and other resources.

If we included the entire \$100 as the farmer's contribution to GDP, it would lead to double counting. This is because the \$40 worth of seeds was already accounted for in the seed industry's output, and including it again would overstate the GDP.

21- A:

The correct answer is A) changes in the average price level.

Personal disposable income per head measures the amount of income available to individuals after taxes and transfers, divided by the population. **However, it does not account for changes in the average price level, also known as inflation.**

In other words, since we don't have **data in "real terms"**, it can be said that changes in the average price level have not been taken into account.

The other options are incorrect because:

B) Personal **disposable** income per head does take into account the level of **direct taxation, as it is calculated after taxes.**

C) Personal disposable income per head is related to national income.

D) The size of the population is explicitly accounted for in the "per head" part of the measure.

22-B:

A – If duties on imports are decreased, there is going to be an increase in consumption of imported goods as they now appear relatively cheaper because of which domestic consumption may actually fall, that translates into a decrease in Aggregate demand. Because of this, production and demand for labor may fall, which results in an increase in demand deficient unemployment.

B – Decreasing unemployment benefits may widen the gap between the take home pay of the employed and the unemployment benefits. This will create incentive effects, encouraging a decrease in **voluntary unemployment**. As a result of which, this becomes correct.

C and D: **Increasing income tax and increasing interest rates** represent contractionary fiscal and monetary policy respectively which are responsible for dampening Aggregate demand, thereby contributing to an **increase in demand deficient unemployment**.

23-A:

Structural Adjustment Programs imposed by the IMF typically aimed to promote a free-market economy and reduce government intervention. Deregulation of agricultural markets would be a key aspect of this. This is why option A is correct.

The other options are inconsistent with SAPs' free-market orientation:

B) Extension of food subsidies would increase government intervention, contrary to SAPs' goals.

C) Nationalization of water supply would expand state control, opposing the privatization and deregulation emphasis of SAPs.

D) Restrictions and regulations on international capital flow would contradict the liberalization and openness encouraged by SAPs.

24-B:

^{old} In a closed economy with no government there is an initial equilibrium level of national income of \$500 billion. ^{a sector multiplier} The economy's households always spend a constant fraction of every \$1 of income they receive. When investment spending increases by \$20 billion the economy moves to a new equilibrium level of national income of \$600 billion. _{new} ΔAE

What is the marginal propensity to consume of the economy's households?

A 0.20 ☒ B 0.80 C 0.83 D 1.00

• change in AE $\rightarrow$ \$20 bn

• change in NI $\rightarrow 600 - 500 \Rightarrow 100$

• multiplier = $\frac{\Delta NI}{\Delta AE} \rightarrow \frac{100}{20} = 5$

multiplier $\Rightarrow$
 $\frac{1}{1 - mpc} = 5$
 $\downarrow$ determine x

$1 = 5 - 5x$
 $1 - 5 = -5x$
 $-4 = -5x$

$\frac{4}{5} = x; x = \frac{4}{5}$
 $\frac{4}{5} = 0.8$

25-C: The transactionary motive depends on personal income level because:

- Higher income individuals have more funds for daily transactions, or to purchase goods and services.

The precautionary motive also depends on personal income level because:

- Higher income individuals may hold more money as a precaution against unexpected expenses or financial shocks, as they have more .

The speculative motive, on the other hand, depends on the level of interest rates because:

- Lower interest rates make holding money more attractive, as the opportunity cost of holding cash is lower.

26-D:

Keynesians assume that:

- Wages are sticky **or inflexible**, meaning they do not adjust quickly to changes in market conditions, such as unemployment or inflation
- By assuming investment is **interest inelastic**, Keynesians highlight the importance of other factors in driving **investment** activity. This contrasts with **classical economics, which assumes investment is highly responsive to interest rates**.

27-B:

Here's a brief explanation of each characteristic:

- High government debt-to-GDP ratio: Many developing economies struggle with high levels of government debt, which can limit their ability to invest.
- Low average propensity to save: Developing economies often have low savings rates, which can limit domestic investment and make them reliant on foreign capital.

28-D:

The correct answer is D) tied aid.

Tied aid refers to development assistance that is conditional on the recipient country. In this case, the US government is providing aid to African countries to fight AIDS/HIV, but requires them to purchase expensive drugs from the US, rather than cheaper alternatives from other countries like South Africa, India, or Brazil.

This is an example of tied aid because the aid is conditional on the recipient country purchasing specific products from the donor country, rather than allowing them to use the funds to purchase the most cost-effective or suitable products.

29-C:

Economists distinguish between economic development and economic growth because:

- **Economic growth** refers to an increase in a country's production capacity, typically measured by GDP. It focuses on the **quantity of output**.
- **Economic development**, on the other hand, encompasses not only economic growth but also improvements in the quality of life, such as:
 - Increased access to education and healthcare
 - Reduced poverty and income inequality
 - Improved infrastructure and institutions
 - Enhanced human capital and technological progress

This explains why key C is correct as **economic development primarily focuses on the concern for quality of life rather than the quantity of output.**

30-B:

B) In an economy operating below full employment, an increase in government expenditure will lead to an increase in aggregate demand, which will stimulate economic activity and increase real national income.

A fall in the propensity to import represents that there is a decrease in withdrawals from the circular flow, and subsequently a rise in domestic consumption as now a greater proportion of the additional income earned will be spent on domestic goods. This will therefore boost Aggregate demand. **This is why, key B, becomes correct.**

A) A fall in taxation may lead to an increase in disposable income and increased consumption, but a rise in the saving rate will reduce consumption and aggregate demand.

C) A rise in investment expenditure will increase aggregate demand, but a rise in taxation will reduce disposable income and subsequently lead to a fall in consumption and aggregate demand.

D) A rise in the propensity to save will reduce consumption and aggregate demand, while a rise in export revenue will boost Aggregate demand and increase national income since it **represents an injection into the circular flow.**

